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global_data = {

# — ATTRIBUTE IDENTITIES
"attribute_1": "total_cost",
"attribute_2": "material_continuity",
"attribute_3": "conductor_size_mm2",
"attribute_4": "voltage_rating_kv",
"attribute_5": "total_run_length_km",

# — ATTRIBUTE DIRECTIONS
"direction_attribute_1": "negative",    # lower cost is better
"direction_attribute_2": "positive",    # higher continuity with historical material is better
"direction_attribute_3": "positive",    # larger conductor → lower resistance → better yield
"direction_attribute_4": "positive",    # higher voltage rating → better safety margin
"direction_attribute_5": "negative",    # shorter run → lower project complexity and cost

# — NORMALIZATION BOUNDS
"attribute_1_min": 3.00,    # $/m — floor: small LV Al cable, lowest-cost market (
"attribute_1_max": 100.00,    # $/m — ceiling: large Cu MV cable, premium certified m

"attribute_2_min": 0,    # binary — previously aluminum
"attribute_2_max": 1,    # binary — previously copper

"attribute_3_min": 16,    # mm² — smallest practical underground solar conductor
"attribute_3_max": 1000,    # mm² — largest commercially produced IEC MV conductor

"attribute_4_min": 0.6,    # kV — LV string/combiner runs lower bound
"attribute_4_max": 66,    # kV — emerging large-project MV collection upper bound

"attribute_5_min": 1,    # km — small community solar / C&I project lower bound
"attribute_5_max": 500,    # km — multi-GW mega-park upper bound

# — COM
"copper_price_per_ton": 9500,
"aluminum_price_per_ton": 2500,
"copper_price_per_kg": 9.500,    # fixed global Cu price
"aluminum_price_per_kg": 2.500,    # fixed global Al price
}

regional_data = [
{
# — UNITED STATES
# Strong copper tradition from NEC heritage; aluminum adoption growing
# but still minority. Balanced weighting with meaningful material
# continuity influence.

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"region":          "United States",
"copper_product_market_share": 0.65,
"aluminum_product_market_share": 0.35,
"weight_attribute_1": 0.28, # total_cost
"weight_attribute_2": 0.18, # material_continuity
"weight_attribute_3": 0.22, # conductor_size_mm2
"weight_attribute_4": 0.17, # voltage_rating_kv
"weight_attribute_5": 0.15, # total_run_length_km
},
{
# — CHINA —————
# Aggressive cost optimization at gigawatt scale; aluminum pragmatism
# dominant; material history carries little weight; large projects
# make run length highly consequential.
"region":          "China",
"copper_product_market_share": 0.40,
"aluminum_product_market_share": 0.60,
"weight_attribute_1": 0.35, # total_cost
"weight_attribute_2": 0.10, # material_continuity
"weight_attribute_3": 0.22, # conductor_size_mm2
"weight_attribute_4": 0.15, # voltage_rating_kv
"weight_attribute_5": 0.18, # total_run_length_km
},
{
# — INDIA —————
# Sub-$0.02/kWh auction pricing drives extreme cost sensitivity;
# youngest utility solar market so historical material preference
# carries the least weight of all four regions.
"region":          "India",
"copper_product_market_share": 0.35,
"aluminum_product_market_share": 0.65,
"weight_attribute_1": 0.40, # total_cost
"weight_attribute_2": 0.08, # material_continuity
"weight_attribute_3": 0.20, # conductor_size_mm2
"weight_attribute_4": 0.14, # voltage_rating_kv
"weight_attribute_5": 0.18, # total_run_length_km
},
{
# — EUROPE —————
# Standards-first procurement (IEC 60364, EN 50618, CPR); technical
# precision and certification compliance dominate; copper tradition
# strong; shorter average project runs reduce run length weight.
"region":          "Europe",
"copper_product_market_share": 0.60,

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    "aluminum_product_market_share": 0.40,
    "weight_attribute_1": 0.22, # total_cost
    "weight_attribute_2": 0.18, # material_continuity
    "weight_attribute_3": 0.25, # conductor_size_mm2
    "weight_attribute_4": 0.23, # voltage_rating_kv
    "weight_attribute_5": 0.12, # total_run_length_km
  },
]

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product_data = [
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{
  "region": "United States",
  "dominant material": "cu",
  "product": "MV Underground Solar Array Collection Cable "
             "– Cu Dominant (240mm² Cu, XLPE, 15kV, Direct Burial)",

  # — MASS BREAKDOWN (per meter) —————
  # 240mm² conductor:  $240\text{e-}6 \text{ m}^2 \times 8,960 \text{ kg/m}^3 = 2.15 \text{ kg/m}$  conductor
  # Cu wire shield: +0.15 kg/m
  # XLPE + HDPE jacket: remaining ~0.40 kg/m (non-metallic, in non_material_cost)
  "copper_kg": 2.30,
  "aluminum_kg": 0.00,
  "total_mass_kg": 2.70,

  # — COSTS (per meter) —————
  # Includes: XLPE/HDPE extrusion, manufacturing labor, NEC compliance
  # testing, termination hardware, trenching & pulling labor (US rates)
  "non_material_cost_per_unit": 14.00, # $/m

  # — ATTRIB
  "attribute_2_value": 1, # material_continuity — previously copper
  "attribute_3_value": 240, # mm²
  "attribute_4_value": 15, # kV
  "attribute_5_value": 120, # km — canonical ~150 MW US utility project
},

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{
  "region": "United States",
  "dominant material": "al",
  "product": "MV Underground Solar Array Collection Cable "
             "– Al Dominant (400mm² Al, XLPE, 15kV, Direct Burial)",

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— MASS BREAKDOWN (per meter) —————

400mm² Al conductor: $400 \times 10^{-6} \text{ m}^2 \times 2,700 \text{ kg/m}^3 = 1.08 \text{ kg/m}$

Cu wire shield (retained in all MV cable designs): 0.15 kg/m

XLPE + HDPE jacket (slightly thicker for larger OD): ~0.52 kg/m

"copper_kg": 0.15,

"aluminum_kg": 1.08,

"total_mass_kg": 1.75,

— COSTS (per meter) —————

\$0.50/m premium over Cu variant: larger cable OD requires more

insulation/jacket material; Al termination hardware slightly costlier

"non_material_cost_per_unit": 14.50, # \$/m

— ATTRIBUT

"attribute_2_value": 0, # material_continuity — previously aluminum

"attribute_3_value": 400, # mm²

"attribute_4_value": 15, # kV

"attribute_5_value": 120, # km

},

{

"region": "China",

"dominant material": "cu",

"product": "MV Underground Solar Array Collection Cable "

"— Cu Dominant (240mm² Cu, XLPE, 15kV, Direct Burial)",

"copper_kg": 2.30,

"aluminum_kg": 0.00,

"total_mass_kg": 2.70,

Vertically integrated Chinese EPC model; low extrusion/labor cost;

GB/T compliance testing included

"non_material_cost_per_unit": 7.00, # \$/m

"attribute_2_value": 1, # material_continuity — previously copper

"attribute_3_value": 240, # mm²

"attribute_4_value": 15, # kV

"attribute_5_value": 200, # km — canonical ~300 MW Chinese utility project

},

{

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"region":          "China",
"dominant material":  "al",
"product":         "MV Underground Solar Array Collection Cable "
                  "– Al Dominant (400mm2 Al, XLPE, 15kV, Direct Burial)",

"copper_kg":       0.15,
"aluminum_kg":     1.08,
"total_mass_kg":   1.75,

"non_material_cost_per_unit": 7.50,  # $/m

"attribute_2_value": 0,    # material_continuity — previously aluminum
"attribute_3_value": 400,  # mm2
"attribute_4_value": 15,   # kV
"attribute_5_value": 200,  # km
},

{
  "region":          "India",
  "dominant material":  "cu",
  "product":         "MV Underground Solar Array Collection Cable "
                    "– Cu Dominant (240mm2 Cu, XLPE, 15kV, Direct Burial)",

  "copper_kg":       2.30,
  "aluminum_kg":     0.00,
  "total_mass_kg":   2.70,

  # Lowest non-material cost: Indian labor rates, IEC-referenced (not
  # third-party-enforced) compliance, competitive local EPC market
  "non_material_cost_per_unit": 6.00,  # $/m

  "attribute_2_value": 1,    # material_continuity — previously copper
  "attribute_3_value": 240,  # mm2
  "attribute_4_value": 15,   # kV
  "attribute_5_value": 160,  # km — canonical ~250 MW Indian utility project
},

{
  "region":          "India",
  "dominant material":  "al",
  "product":         "MV Underground Solar Array Collection Cable "
                    "– Al Dominant (400mm2 Al, XLPE, 15kV, Direct Burial)",

```

"copper_kg": 0.15,
"aluminum_kg": 1.08,
"total_mass_kg": 1.75,

"non_material_cost_per_unit": 6.50, # \$/m

"attribute_2_value": 0, # material_continuity — previously aluminum
"attribute_3_value": 400, # mm²
"attribute_4_value": 15, # kV
"attribute_5_value": 160, # km

},

{

"region": "Europe",
"dominant material": "cu",
"product": "MV Underground Solar Array Collection Cable "
"— Cu Dominant (240mm² Cu, XLPE, 15kV, Direct Burial)",

"copper_kg": 2.30,
"aluminum_kg": 0.00,
"total_mass_kg": 2.70,

Highest non-material cost: EU labor rates, CE marking, EN 50618 /

IEC 60502 third-party certification (TÜV/DNV), CPR compliance,

stringent permitting and environmental documentation

"non_material_cost_per_unit": 15.00, # \$/m

"attribute_2_value": 1, # material_continuity — previously copper
"attribute_3_value": 240, # mm²
"attribute_4_value": 15, # kV
"attribute_5_value": 50, # km — canonical ~80 MW European utility project

},

{

"region": "Europe",
"dominant material": "al",
"product": "MV Underground Solar Array Collection Cable "
"— Al Dominant (400mm² Al, XLPE, 15kV, Direct Burial)",

"copper_kg": 0.15,
"aluminum_kg": 1.08,

"total_mass_kg": 1.75,

"non_material_cost_per_unit": 15.50, # \$/m

"attribute_2_value": 0, # material_continuity — previously aluminum

"attribute_3_value": 400, # mm²

"attribute_4_value": 15, # kV

"attribute_5_value": 50, # km

},

]